

Goal Parity

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KEY FINDINGS

- All assets and liabilities in any portfolio should be thought of as means contributing to the following four ends: *Liquidity, Income, Preservation, Growth*.
- Assets contributing to each of the four goals perform best in different environments, so diversification can be better achieved if all the four goals are “powered.”
- Over 50+ years of simulated performance, our Goal Parity portfolios achieve highly competitive risk-adjusted returns relative to traditional benchmarks.

ABSTRACT

Following “4x4 Asset Allocation” by Cron and Golts (2022), all assets and liabilities in any portfolio should be thought of as means contributing to the following four ends:

- *Liquidity* maintenance: nominally safe and quickly accessible “cash-like” pool of assets
- *Income* generation: relatively regular, certain and near-term cash payments
- *Preservation* of (real) capital: assets expected to retain their value over time
- *Growth*: more volatile assets and strategies expected to generate long-run returns.

We illustrate the investor-specific quantification of these four goals with decades of empirical data for a wide variety of liquid and private assets, fund indexes, and alternative premium strategies. Further, using 1970–2022 data, we simulate strategic Goal Parity Balanced portfolios, seeking to have all four goals equally “powered,” as well as strategic Goal Tilted portfolios emphasizing one of the goals. Over the 50+ years of our simulation, Goal Parity portfolios achieve highly competitive risk-adjusted returns relative to traditional benchmarks, in particular delivering shallower drawdowns in various crisis and inflationary periods. Our Growth-tilted portfolio diversifies and outperforms equities with shallower drawdowns, and our Preservation-tilted portfolio has a very distinct performance pattern: it lags equities during bull markets but delivers robust performance in bear markets. Overall, we empirically illustrate a powerful goal-based, investor-specific asset management framework.

In “4x4 Asset Allocation,” Cron and Golts (2022) argued that all assets and liabilities in any portfolio should be thought of as means contributing to the following four ends:

- *Liquidity* maintenance: nominally safe and quickly accessible “cash-like” pool of assets,
- *Income* generation: relatively regular, certain and near-term cash payments,

- *Preservation* of (real) capital: assets expected to retain their value over time, and
- *Growth*: more volatile assets and strategies expected to generate long-run returns.

The four goals are schematically depicted in **Exhibit 1**. In this article we provide empirical evidence supporting the idea that the natural diversification arising from paying attention to all four goals should allow a portfolio to perform in all four “terrains” or market environments, stylistically described as recession, recovery, mid-cycle, and late-cycle.

Further following Cron and Golts (2022), we suggest that investors should start their asset allocation process by setting a *strategic investment horizon*, T , over which they would seek to achieve their goal—for example, 3, 10, or 30 years. Investors then should set a *substantial loss barrier* B , for example 95%, 85%, or 70%, indicating their “pain” threshold. For example, investors with $B = 85\%$ can tolerate losing up to $(1 - B) = 15\%$ of their portfolio at any point over the strategic horizon, but would be very averse to losses greater than that. In one practical application, such a barrier B can be driven by the investor’s (estimated) liabilities.

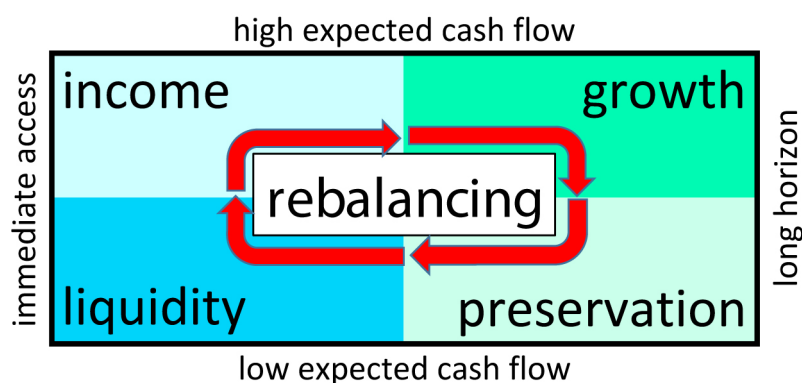
An investor interested in a balanced portfolio may seek Goal Parity, i.e., have roughly equal strategic Liquidity, Preservation, Income, and Growth allocations. Should the investor have active views, for example on the cyclical or secular economic environment, our construction allows for overweighting of one or more of the goals, creating Goal Tilted portfolios: Liquidity, Income, Preservation, and Growth.

The next step in the asset allocation process should be *tactical rebalancing*. A portfolio should be reviewed and rebalanced with some regular frequency τ , with each tactical rebalancing seeking to re-align the portfolio with the strategic goals. The tactical rebalancing process incorporates (tactical) alphas, covariance matrixes, constraints and transaction costs that we do not investigate explicitly in this article. But we illustrate the process schematically in Exhibit 2.

The article is organized as follows. First, we analyze several dozen traditional and alternative assets starting from the familiar mean-variance efficient frontier, discuss the relationship between Sharpe ratio and skewness of returns, introduce the concepts of “defaultility” and “liquility” by adjusting volatility for skewness, and arrive at the Growth, Income, Preservation, Liquidity—abbreviated as GIPL—Asset

EXHIBIT 1

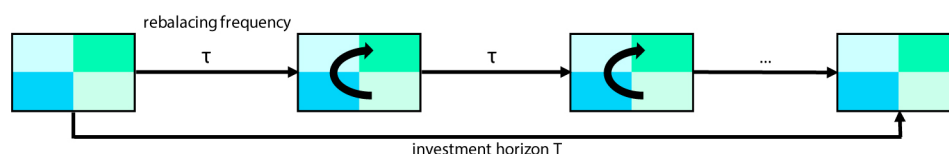
Four Investment Goals



NOTE: Time horizon and cash flow heuristics of the investments associated with four investment goals.

EXHIBIT 2

Rebalancing over the Strategic Horizon



NOTES: Strategic investment horizon T (for example, 1, 3, 10, or 30 years), and tactical rebalancing frequency τ (for example, daily, weekly, monthly, or quarterly). We suggest that investors should first build strategic portfolios, and then rebalance allocations on a tactical frequency, re-aligning portfolios with the strategic goals and managing tactical risk and return, and to ensure a minimum liquidity position and cash flow.

Map perspective. We then perform simulations of the goal-based strategic portfolios with an eight-asset universe, constructing Goal Parity as well as four Goal Tilted portfolios that compare favorably with traditional and hedge fund index benchmarks and may be of practical use to investors. Further, we illustrate the customizability of goal-based portfolios to investor preferences. Finally, we discuss a perspective on Strategic Asset Allocation as tracking nominal GDP after consumption. In the Appendixes, we apply our methodology to global and “endowment” universe assets, and Fama-French factor portfolios. The Appendixes also provide additional information on data sourcing and methodology.

RELATED LITERATURE AND INDUSTRY APPROACHES

The quantitative asset allocation introduced by Modern Portfolio Theory is more than 70 years old now, and the literature on asset allocation and portfolio construction is vast and fast-growing (we would refer the reader to several recent books, such as Elton et al. (2014); Page (2020); Kinlaw, Kritzman, and Turkington (2021); and references therein).

Approaches to strategic asset allocation include the 60/40 equity/bond allocation, modifications of 60/40 including substantial allocation to alternatives and private assets, market cap-based and fundamental benchmarks for local and global markets, forecast-based benchmark-free allocation, and optimized approaches including risk parity as well as higher-moment-aware optimization (see, for example, Harvey and Siddique 2000 and 2022; Harvey et al. 2010, and Lezmi et al. 2018). Very long horizon institutional investors may follow the “endowment” model investing in liquid as well as private, illiquid sectors (Ang et al. 2020; Shepard, Qiu, and Li 2022). Strategic Asset Allocation is studied in Campbell and Viceira (2002).

Investor preferences in theory and practice are often one-dimensional: “risk aversion” parameter in mean-variance optimization, “conservative” to “strategic” sliding scale, or a strategic horizon (years to the target date) in target date funds utilizing a “glide path” approach to equity/bond allocation.

The goal-based investment and wealth management approach has been explored by academics and practitioners alike. For a recent overview of Liability-Driven Investment (LDI) see Tamas-Hastings (2021). Wilcox, Horvitz, and diBartolomeo (2006) relate ratio of total assets to discretionary wealth with risk aversion. Merton and Muralidhar (2020) provide an extensive literature review, and discuss income, investment preferences, and goal-based approaches in the context of a looming retirement crisis.

The 4x4 Asset Allocation introduced in Cron and Golts (2022) is a goal-based framework explicitly incorporating investor-specific loss aversion and strategic horizon

preferences at its first, strategic stage, and then utilizing mean-variance optimization techniques at a second, tactical stage. In this article we focus on empirically investigating the goal-based Growth, Income, Preservation, Liquidity—abbreviated as GIPL—strategic allocation.

QUANTIFYING THE FOUR GOALS

In this section we define and illustrate the quantification of the Growth, Income, Preservation, and Liquidity (GIPL) goals.

Data

We assemble a universe of (mostly) US-centric assets from March 2000 to December 2022, including major equity indexes such as S&P 500, Nasdaq-100, MSCI US Equity Indexes (Value, Growth, Small Caps, etc.), Bloomberg Fixed Income Indexes (US Aggregate, Treasuries, Investment Grade credit, High Yield, TIPS, etc.), HFRI hedge fund strategies, several Commodity Trading Advisor¹ (CTA) indexes, gold, as well as some long- and short-volatility strategies. More details on the data are presented in Appendix D: Data Details, which also contains a panel of global assets. All data is monthly. All subsequent calculations are performed by the authors.

Building the GIPL Map: Traditional and Alternative Assets

In this section we illustrate the quantification of the four goals using the universe and data described above.

We start with the traditional efficient frontier of stocks (represented by S&P 500 Total Return Index) and bonds (represented by Bloomberg US Agg Total Return Index). This efficient frontier is depicted on the return vs. volatility chart in Exhibit 3, alongside other traditional equity and bond assets, as well as hedge fund indexes, CTA, trend and “long volatility” strategies, and gold. We would like to look at this chart from an investor’s goal-based perspective with the following intuition:

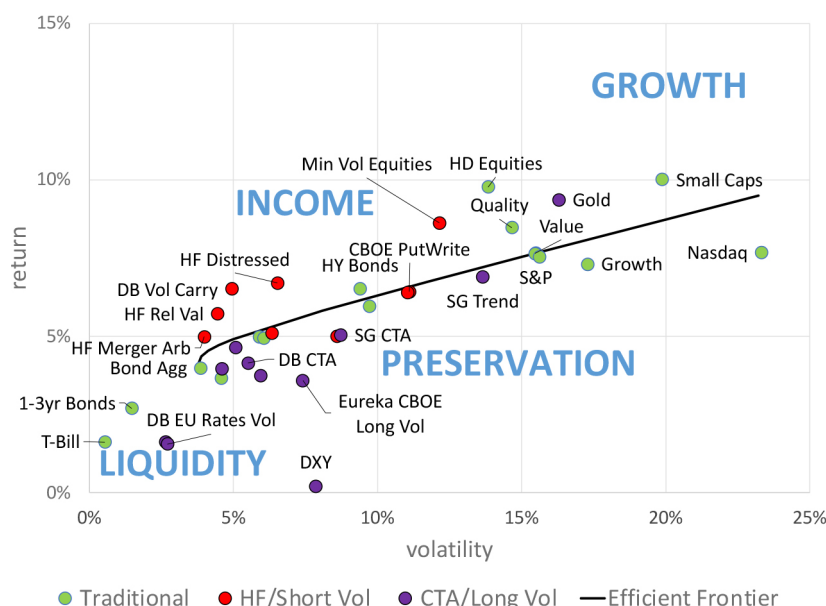
- Volatile equities with expected higher future cash payments contribute more to *Growth*.
- High yield bonds and hedge fund-style managed strategies are implicitly “short volatility” and tend to be above our traditional efficient frontier with higher Sharpe ratio, contributing more to the *Income* goal.
- CTA and “long volatility” managed strategies tend to be below our traditional efficient frontier with lower Sharpe ratio, contributing more to the longer-term *Preservation* goal.
- Shorter duration bond and T-bills contribute to shorter-term preservation or *Liquidity* goal.

The concept of risk is complicated and multi-dimensional. In financial theory and practice an asset’s “risk” is often summarized by—and sometimes used interchangeably with—the asset’s volatility estimate. However, financial returns are often asymmetric, with potentially sharp losses, or in some cases potentially sharp gains.

¹ Strategies implemented by firms classified as “Commodity Trading Advisors” for regulatory and historical reasons often employ momentum and other technical signals for bond, interest rate, equity, and commodity futures, as well as FX forwards and other instruments.

EXHIBIT 3

Traditional Efficient Frontier



NOTES: Period: 2000–2022. Traditional equities (S&P 500 Total Return Index) and bonds (Bloomberg US Agg Total Return Index) efficient frontier chart for illustration and educational purposes only. The x-axis “volatility” is computed as annualized standard deviation of monthly returns; the y-axis “return” is computed as arithmetic average of monthly returns.

The “long/short volatility” intuition for assets and strategies suggests the following tradeoff between Sharpe ratio and returns asymmetry:

- Higher Sharpe ratio assets/strategies tend to have negative skewness, and contribute more to the Income goal.
- Lower Sharpe ratio assets/strategies tend to have positive skewness and be “scarce,”² contributing more to the Preservation goal.

Indeed, such a tradeoff is empirically confirmed in literature using various asymmetry and skewness metrics (see Engle 2011; Lezmi et al. 2018; Lempérière et al. 2017; Baltas and Salinas 2022, and Acadian Asset Management 2020). We report the relationship between Sharpe ratio and skewness using our 2000–2022 data in Exhibit 4.

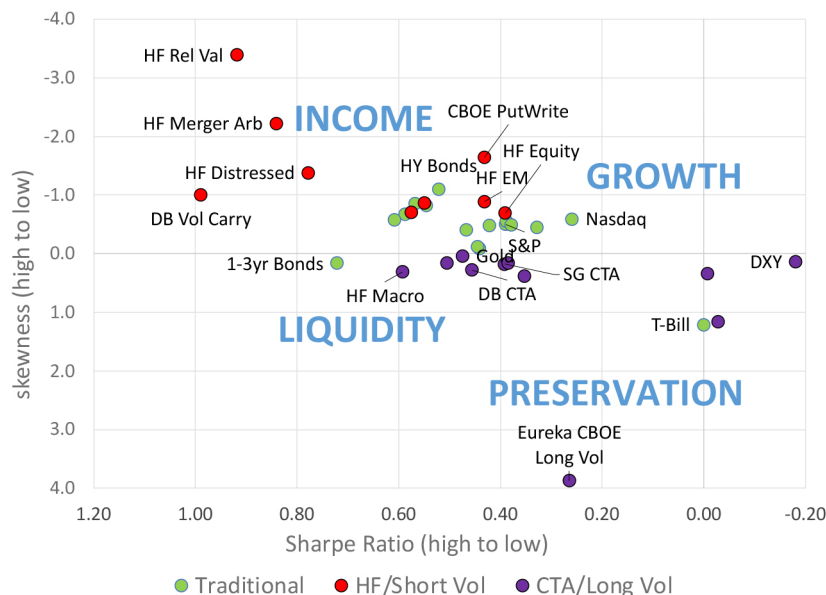
We observe that a measure of return asymmetry is required to fully access risk. There exists a substantial and deep literature on asset pricing and portfolio construction with higher moments (see Harvey and Siddique 2000, 2022; Harvey et al. 2010, and references therein). Including the effect of skew and kurtosis to compute an asset’s “volatility equivalent” is discussed in diBartolomeo (2022/2023) and references therein.

Further, we know that investors bear more than one risk, including the general notions of default risk and liquidity risk. Thus, our risk modeling approach, described in Appendix A: Skewness and Horizon-Driven Liquidity Preference, is to define two

²Positively (resp. negatively) skewed assets tend to be “scarce” and convex (resp. “productive” and concave) with respect to major risk factors including interest rates, inflation, and credit spreads (see Golts and Jones 2024).

EXHIBIT 4

Sharpe Ratio and Skewness



NOTES: Period: 2000–2022. The x-axis “Sharpe Ratio” is computed with monthly returns in excess of T-bills; the y-axis “skewness” is Fisher’s moment coefficient of skewness (third standardized moment) computed with monthly returns.

aspects or components of risk, accounting for assets’ skewness γ to adjust volatility σ as follows:³

$$\sigma_D = \sigma \exp\left(-\frac{\gamma}{2\gamma_0}\right) \text{ and } \sigma_L = \sigma \exp\left(\frac{\gamma}{2\gamma_0}\right).$$

The adjusted volatilities σ_D and σ_L are dubbed “defaultility” and “liquidity” respectively, see Exhibit 5. Intuitively:

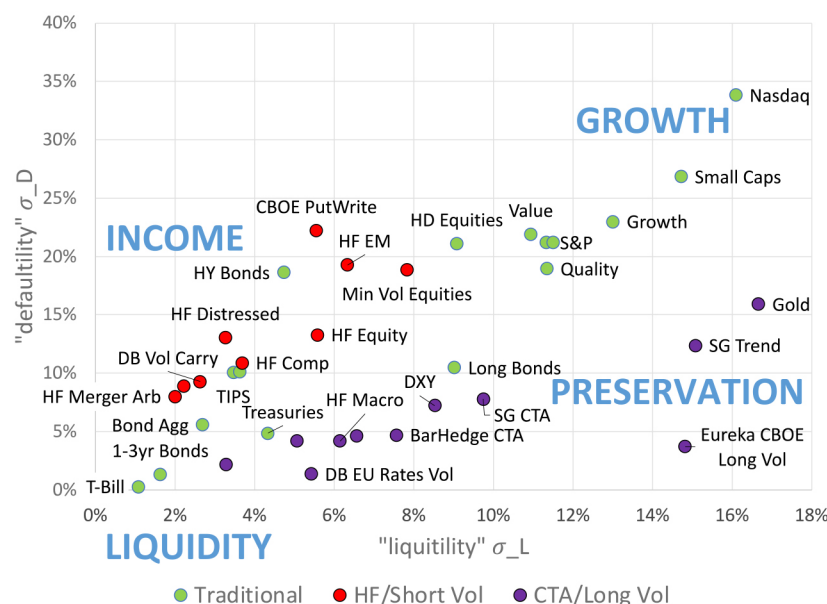
- **Defaultility** reflects left-tail risk, such as a negatively skewed asset’s default or shortfall, potentially imperiling the return of capital.
- **Liquidity** reflects right-tail risk, with a potentially lower mean or interim return on capital while waiting for a positively skewed asset’s embedded “lottery ticket” to pay off.

Typically, the “defaultility” risk is more short-to-medium term (e.g., the default risk on fixed-maturity corporate bonds), whereas “liquidity” risk refers to assets whose valuation comes from more distant expected cash flows. Heuristically, long-horizon assets, as well as real assets with an inflation-protected nature, enjoy higher convexity relative to short-term and nominal assets, consistent with our framework. Just like in our description of the Liquidity goal, the “liquidity” risk here refers to whether assets can reliably serve the role of returning capital to the investor when needed. Indeed, Keynes (1964; orig. pub. 1936) referred to the preference for short-term over long-term assets as a “Liquidity preference.” Tédongap and Tafolong (2017) observe that

³ Here, γ_0 is a parameter related to the fat-tailed nature of the returns. We use $\gamma_0 = 1.6$ in this article.

EXHIBIT 5

“Defaultity” and “Liquidity”



NOTES: Period: 2000–2022. The x-axis “liquidity” is higher (resp. lower) than volatility for positively (resp. negatively) skewed assets. The y-axis “defaultity” is lower (resp. higher) than volatility for positively (resp. negatively) skewed assets.

“illiquidity is an important dimension of investment decisions that is complementary to the traditional mean-variance framework.”

Now we would like to introduce investor preferences and use “defaultity” σ_D and “liquidity” σ_L to quantify the investor-specific contributions of all assets and strategies to the Growth, Income, Preservation, and Liquidity goals. Recall the investor-specific parameters introduced above:

- T : strategic investment horizon, say, 10 years.
- B : “substantial loss” barrier, very averse to losing, say, more than $(1 - B) = 15\%$ of the portfolio’s net worth.
- τ : tactical rebalancing/trading frequency, say, a quarter or more frequently.⁴

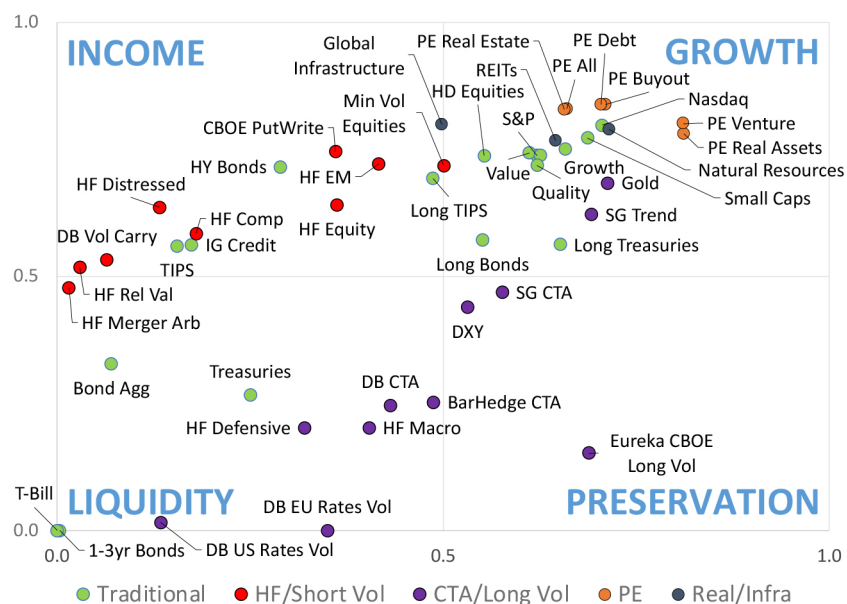
The GIPL asset map represents the universe of assets in a $[0, 1] \times (0, 1)$ unit square, where Liquidity, Income, Preservation, and Growth goals each correspond to the four corners. Heuristically, the vertical side of the square refers to the “defaultity” risk and the horizontal side refers to the “liquidity” risk. The positioning of each asset in the square is determined by its risk characteristics and the investor’s preferences; the procedure is described in Appendix B: GIPL Goal Map.

Assuming a strategic investment horizon $T = 10$ years and $B = 0.85$, the contributions of our assets to the Growth, Income, Preservation, and Liquidity goals are computed and displayed in Decomposition of the Asset Contributions to the Four Goals in Appendix D: Data Details. We then depict the assets’ goal decomposition in Exhibit 6. Here, the x-coordinate of an asset-representing dot is the sum of its contribution to Growth and Preservation, and the y-coordinate is the sum of its contribution to Growth and Income (see Appendix B: GIPL Goal Map). Heuristically, the greater the x-coordinate, the longer is the asset’s “horizon” and thus (investor-specific) “liquidity risk”

⁴We assume small values $\tau \rightarrow 0$, which means investors expect to potentially perform frequent rebalancing, as well as access strategy for liquidity as needed.

EXHIBIT 6

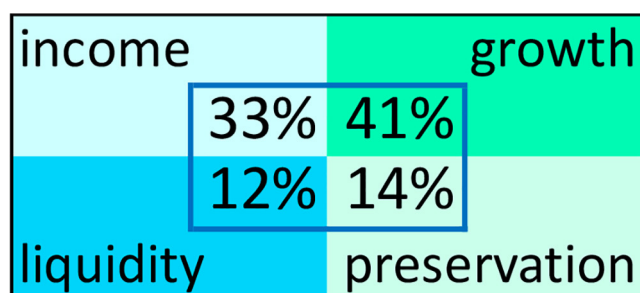
GIPL Asset Map: Investor's Goals



NOTES: Period: 2000–2022. 4x4invest's proprietary methodology, strategic horizon $T = 10$ years, "substantial loss" barrier $B = 85\%$.

EXHIBIT 7

Goal Contributions Example: High-Dividend Equities



(Hill 2009), and the greater the y-coordinate the larger is the asset's (investor-specific) "default risk."

Thus we are looking at all assets and strategies from the investor-specific goal-based perspective.

Let us take an example of High Dividend (HD) Equities. Investors whose strategic investment horizon is $T = 10$ years and who are very averse to losses beyond $B = 85\%$ of portfolio's net worth would see HD equities contributing to the four goals as in Exhibit 7. If investors were oriented toward growth and income, they would tend to like High Dividend equities because these contribute 41% to the investor's Growth goal, 33% to the Income goal, 14% to the Preservation goal, and 12% to the Liquidity goal. (This decomposition, of course, depends on the period-specific estimates of expected volatility and skewness, and on market conditions insofar as they influence volatility and skewness estimates.)

Discussion: Hedge Funds and CTAs

Negatively skewed HFR indexes (HFRI Merger Arbitrage Index, HFRI Distressed/Restructuring Index, HFRI Relative Value Total Index, HFRI Equity Hedge Total Index, HFRI Emerging Markets Total Index, and HFRI Fund Weighted Composite Index) share the Income "corner" with High Yield Bonds on the GIPL Asset Map. By contrast, HFRI Macro Total Index and HFRI FOF Market Defensive Index are closer to the CTA indexes (see Appendix D: Data Details). The GIPL classifications also are supported from the point of view of standard correlations. From the monthly correlations' viewpoint, the negatively skewed HFR indexes are 50%–80% correlated with High Yield Bonds and S&P 500 Index, as well as each other (see Exhibit 8).

EXHIBIT 8

Hedge Fund Correlations

Assets	HF Merger Arb	HF Distressed	HF Rel Val	HF Comp	HF Equity	HF EM	HF Macro	HF Defensive	BarCap CTA
HF Merger Arb	100%	63%	63%	69%	64%	53%	32%	24%	-2%
HF Distressed	63%	100%	82%	81%	74%	70%	38%	24%	-7%
HF Rel Val	63%	82%	100%	77%	73%	65%	33%	28%	-7%
HF Comp	69%	81%	77%	100%	96%	85%	61%	40%	5%
HF Equity	64%	74%	73%	96%	100%	76%	51%	35%	3%
HF EM	53%	70%	65%	85%	76%	100%	52%	33%	-3%
HF Macro	32%	38%	33%	61%	51%	52%	100%	63%	49%
HF Defensive	24%	24%	28%	40%	35%	33%	63%	100%	66%
BarCap CTA	-2%	-7%	-7%	5%	3%	-3%	49%	66%	100%

Assets	HF Merger Arb	HF Distressed	HF Rel Val	HF Comp	HF Equity	HF EM	HF Macro	HF Defensive	BarCap CTA
HY Bonds	54%	70%	70%	68%	63%	60%	26%	4%	-11%
S&P	54%	54%	54%	76%	76%	62%	30%	7%	-7%

NOTES: Period: 1990–2022. Standard Pearson correlation coefficients computed with monthly returns.

Negatively skewed HFR indexes are only 20%–50% correlated with HFRI Macro Total Index, 20%–40% correlated with HFRI FOF Market Defensive Index, and uncorrelated with BarclayHedge CTA Index. The CTA Index is also slightly negatively correlated with High Yield Bonds and S&P 500. Put together, this provides correlation-based support for the significant Preservation goal contribution from CTAs, Global Macro, and Defensive strategy funds.

Discussion: Private Equities

According to Pedersen, Page, and He (2014), “return indexes for privately held assets are often artificially smoothed,” so “to adjust for the smoothing effect, our model assumes observed index returns represent a ‘moving average’ of the current and past ‘true’ investment returns.” Czaronis, Kinlaw, and Turkington (2021) argue that private equity volatility is similar to public equity volatility. According to Baz et al. (2022), private equity can be proxied by “Small, Cheap, Leveraged Public Equities” (see also Belev and diBartolomeo 2021). Using similar “un-smoothing” techniques and Small Cap equity proxies, we adjust private equity index volatilities upward and report the results in **Appendix D: Data Details**. Our private equity adjusted volatilities of 25%–35% are in line with the adjusted vols in Pedersen, Page, and He (2014) (private equity: 22%, venture capital: 52%) and Baz et al. (2022): “true economic volatility of private equity is close to 30%, rather than a headline number of 10%.”

Note that, based on quarterly data, Bloomberg Real Assets Private Equity Index and Bloomberg Venture Capital Index have small positive and approximately zero skewness respectively, while the rest of private equity indexes are negatively skewed.

Liquidity risk of private equity must be properly measured and managed in order to realize the potential illiquidity premium. Option-based approaches to liquidity risks of private assets are discussed in particular in Golts and Kritzman (2010); Hayes, Primbs, and Chiquoine (2015), and Lindsey and Weisman (2016). The 4×4 framework uses option pricing theory described in Appendix A: Skewness and Horizon-Driven Liquidity Preference to access liquidity risks. In particular, our asset allocation will not rely on private equity allocation to achieve its Liquidity goal.

Discussion: Reits and Infrastructure

We include real estate in our investment universe by proxying this asset class with Dow Jones Equity REIT Total Return Index. Shaver et al. (2023) consider the relationship of REITs with other real estate investments and their role in asset allocation. We proxy infrastructure with S&P Global Infrastructure Total Return Index. This index is negatively skewed, about 70–80% correlated with REITs, High Dividend Equities, and High Yield bonds, and 30%–40% correlated with TIPS. On the GIPL map in Exhibit 6, S&P Global Infrastructure falls between Growth and Income, a bit more Income-y than REITs and HD Equities, a bit more Growth-y than HY Bonds.

Discussion: Fama-French Factors

In **Appendix C** we look at Fama-French factors over 1963–2022 and observe that—according to our methodology with a 10-year strategic horizon and 15% loss tolerance—Value (HML) has a significant Preservation contribution, providing some protection in recessions, while CMA and LT_Rev factors are even better Preservation factors. In particular HML, CMA, and LT_Rev did very well in 2022 when both stocks and bonds declined.

PORTFOLIO SIMULATION: EIGHT ASSETS, 1970–2022

In this section we investigate the simulated performance of goal-based Strategic Allocation portfolios, investing with a small but representative universe of eight assets, across five decades, see Exhibit 9. First, we describe the process of constructing goal-seeking portfolios using the GIPL Goal Map described previously. Then, we review the data being used for a point-in-time, no look-ahead bias simulation running from 1970 to 2022, thus covering a variety of regimes and economic shocks. The asset parameters for the first simulation are set to achieve “Goal Parity” and the simulation performs favorably relative to some basic benchmarks, achieving long-term growth as well as risk and drawdown control. We then illustrate the flexibility of our framework to the incorporation of “Goal Tilts” (that may perhaps be driven by investor views), as well as to customizability according to investor preferences for horizon and risk.

Strategic goal-based asset allocation portfolios are constructed using an objective function designed to maximize expected return and asset diversification while seeking to achieve goal targets. Let E (Return) be the expected strategic portfolio return, derived from the assets’ expected returns. Then, the investor sets out allocation targets s_L, s_I, s_P, s_G for Liquidity, Preservation, Income, and Growth respectively, where $s_L + s_I + s_P + s_G = 100\%$. The deviation of a portfolio from the goal targets (on a sum-of-squares basis for each goal) is denoted D (Port vs Goals). Then the portfolio is constructed to maximize the following schematic objective function:

$$E(\text{Return}) - D(\text{Port vs Goals}) - \text{Concentration Index}.$$

The last term (concentration index of the portfolio) is a normalizing term that renders the optimization stable and well-behaved, and, *ceteris paribus*, prefers asset diversification. Further technical details can be found in Cron and Golts (2022, section **Asset Allocation: Strategic 4x4 Goal Based Process**). We simulate here a fully invested portfolio by constraining the asset weights to add up to 100%.

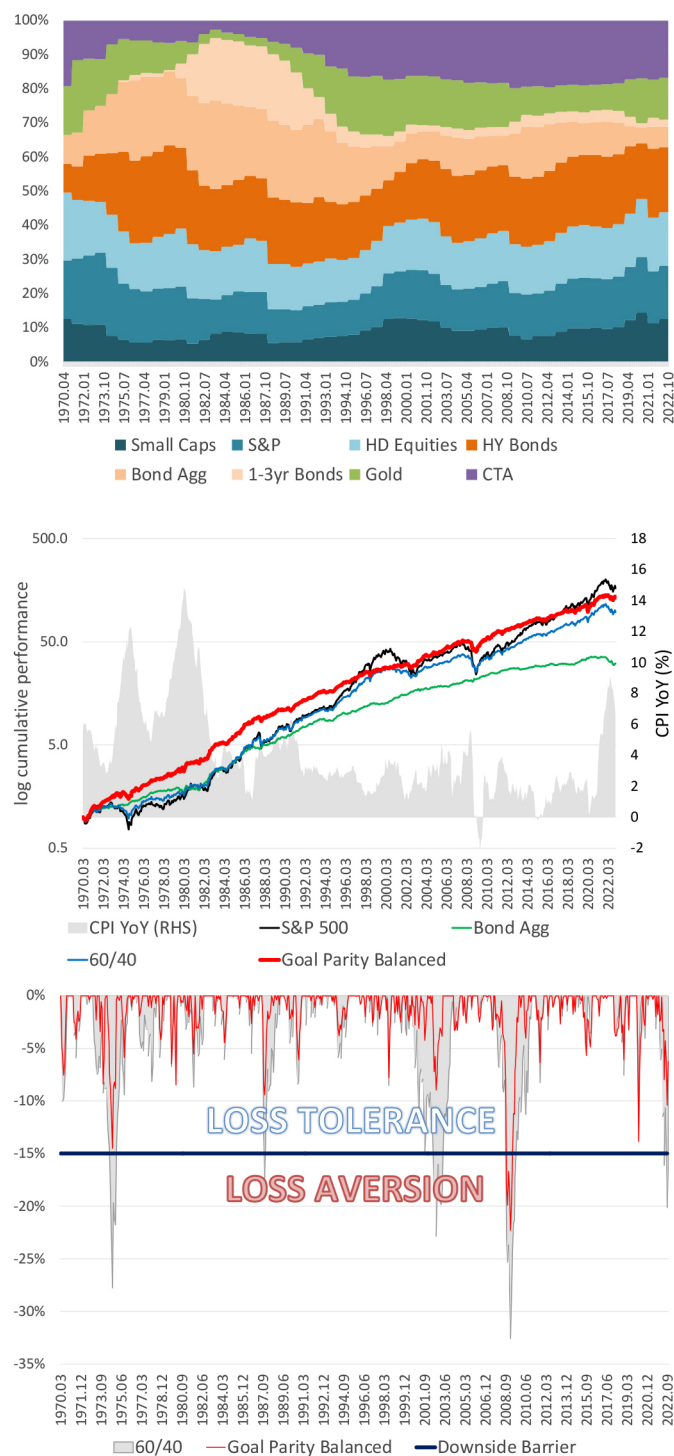
EXHIBIT 9

Monthly Data for a Universe of Eight Assets, March 1970 to December 2022

Short Name	Long Name
Small Caps	Small Cap Equities
S&P	S&P 500 Total Return Index
HD Equities	High Dividend Equities
HY Bonds	Bloomberg US Corp High Yield Total Return Index
Bond Agg	Bloomberg US Agg Total Return Index
1–3 yr Bonds	Bloomberg 1–3 yr Gov/Credit Total Return Index
Gold	Gold Spot price per Troy Ounce
CTA	CTA Index

EXHIBIT 10

Goal Parity Balanced Portfolio Weights, Cumulative Performance, and Drawdown



NOTES: Period: 1970–2022. Strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$. Upper panel depicts weights of eight assets in the portfolio, middle panel depicts cumulative performance (log-scale y-axis), lower panel depicts drawdowns from high watermark. Simulated performance is not indicative of actual future results. For illustration and educational purposes only.

Data and Methodology

We consider monthly data for a universe of eight assets from March 1970 to December 2022. The data is described in detail in Appendix D: Data Details.

We estimate volatility using exponentially (420-month half-life) weighted returns over trailing 120-month window, and skewness using returns Winsorized at $\pm\sigma$ over expanding window. Further, skewness estimates are also Winsorized at ± 1 . We ensure there is no look-ahead bias in simulations: all estimates, in particular of volatility, skewness, and GIPL Goal Maps, are performed with point-in-time data.

We generate portfolios without the use of strategic expected returns, i.e., the strategic expected returns are assumed to be zero. Thus, the portfolios simply seek to achieve the four goal targets, while respecting a preference for diversification.

We perform optimization once a year, at the end of December, and hold the portfolios static for a year (rebalancing monthly to account for market movements). We do not take transaction costs into account for any assets, but their presence would not materially change the performance patterns or statistics due to the infrequent nature of trading.

Goal Parity Balanced Portfolios

The Goal Parity Balanced portfolios place equal weight on all four goals, that is

$$s_L = s_I = s_P = s_G = 25\%.$$

We believe all four goals need to be equally “powered” in a well-diversified portfolio. These goal targets compete with other components of the objective function (and constraints) and may not necessarily be achieved precisely. However, since the deviation of a portfolio from the goal targets term in the objective function imposes just three extra degrees of freedom on the system, the optimizer will get very close in a large and diverse enough asset universe.

We report the weights and cumulative performance of the simulated⁵ Goal Parity Balanced portfolio in Exhibit 10, and performance statistics in Exhibit 11. Note that the Goal Parity Balanced port-

⁵The optimized returns presented in this article are for simulated portfolios. Simulated performance does not represent actual trading or actual trading account but rather is achieved by means of retroactive application of a 4x4invest’s proprietary asset allocation model. Simulated performance is not indicative of actual future results. For illustration and educational purposes only.

EXHIBIT 11

Goal Parity Balanced Portfolio Performance Statistics

1970–2022	S&P	Bond Agg	60/40	Goal Parity Balanced
Mean, annualized	10.9%	6.7%	9.2%	9.63%
Stdev, annualized	15.4%	5.3%	9.9%	7.84%
Worst Month	–21%	–6%	–11%	–11%
Best Month	18%	11%	11%	9%
Sharpe Ratio	0.43	0.44	0.49	0.68
Worst Drawdown	–51%	–17%	–33%	–22%
2002 Drawdown	–24%	–17%	–20%	–10%
2022 Return	–18%	–13%	–16%	–5%
Bear Markets	S&P	Bond Agg	60/40	Goal Parity Balanced
1972.12–1974.09	–45%	7%	–28%	–4%
1987.08–1987.11	–29%	2%	–17%	–9%
2000.08–2002.09	–45%	23%	–23%	–3%
2007.10–2009.02	–51%	6%	–33%	–22%
2021.12–2022.09	–24%	–15%	–20%	–10%

NOTES: Period: 1970–2022. Strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$.

folio returns not far below S&P, but with half the volatility and shallower drawdowns in all five bear market periods we identify since 1970. The Goal Parity Balanced drawdowns are also shallower than the 60/40 portfolio (60% S&P and 40% Bond Agg, rebalanced monthly).

The Goal Parity Balanced portfolio (simulated with the most basic setting) demonstrates the efficacy of the investment framework for investing with a universe of quite varied asset types. The strategy also delivers an ex-post risk and drawdown profile broadly commensurate with the defining risk preferences of the investor. We do not estimate or use correlations for strategic portfolios, in part because multi-year, low-frequency correlations may be quite different from higher frequency (monthly) correlations, and harder to estimate due to data sample limitations (see Kinlaw, Kritzman and Turkington 2014). However, the goal-orientation with its focus on the asymmetry of risks across the spectrum of assets, works to curtail the drawdown of the many observed episodes in our simulations. We suggest to reserve the use of covariance matrix estimates for a second, tactical stage of the portfolio construction process.

Goal-Tilted Portfolios

Our methodology allows orienting (or “tilting”) portfolios to one or more of the Liquidity, Income, Preservation, and Growth goals by upweighting the s_L , s_I , s_P , and s_G allocation targets. We illustrate this flexibility with four simulated portfolios maximizing the tilt to each respective goal within the available asset universe:

- Growth: allocation targets $s_L = s_I = s_P = 0$ and $s_G = 100\%$
- Income: allocation targets $s_L = s_P = s_G = 0$ and $s_I = 100\%$
- Preservation: allocation targets $s_L = s_I = s_G = 0$ and $s_P = 100\%$
- Liquidity: allocation targets $s_I = s_P = s_G = 0$ and $s_L = 100\%$

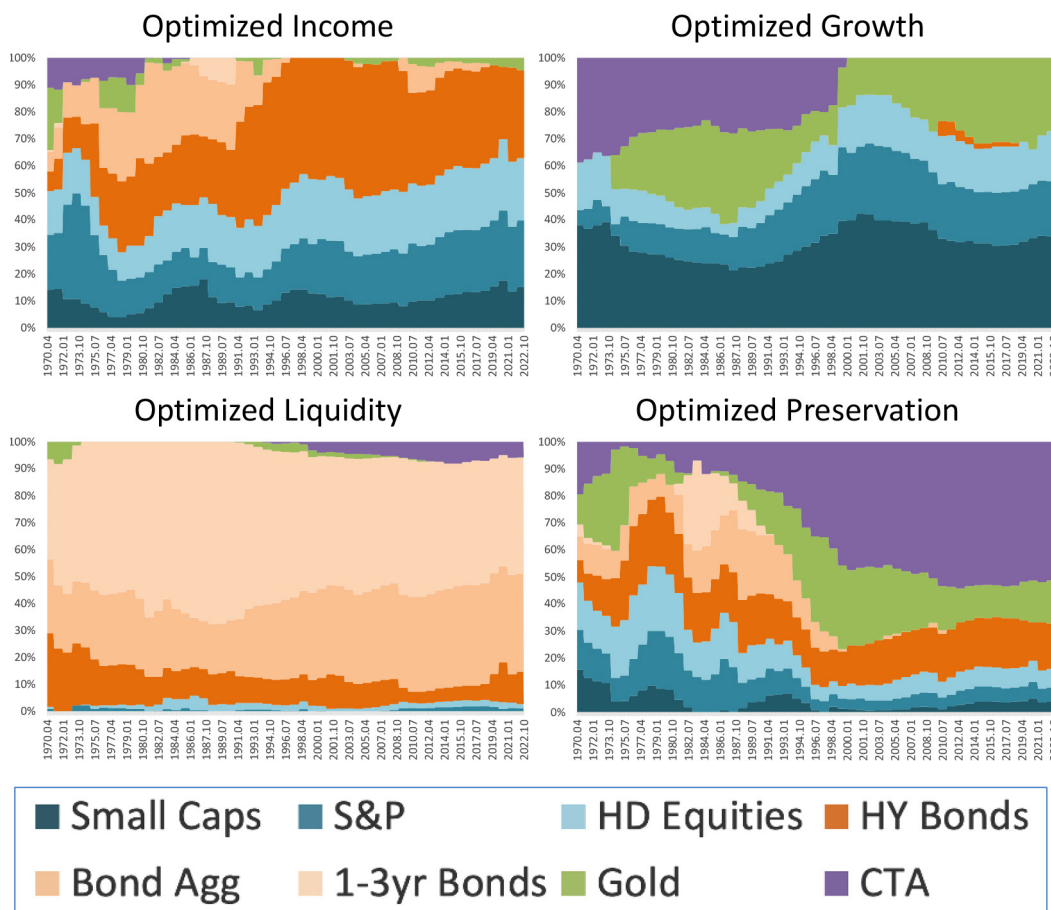
The weights of the four portfolios above are illustrated in Exhibit 12. We observe that:

- Growth has heavy allocation to equities diversified with gold.
- Income allocates to High Yield Bonds and High Dividend Equities.
- Preservation has heavier allocation to CTAs and gold, diversified with High Yield Bonds and equities.
- Liquidity is mostly 1–3 yr bonds and Bond Agg, diversified with High Yield Bonds and CTAs.

We report and depict simulated performance of the four portfolios in Exhibits 13 and 14. Note that Growth portfolio is a diversified portfolio outperforming S&P 500, with drawdowns significantly better than the S&P 500 and handily beating 60/40 on an overall basis. The Preservation portfolio has a very distinct performance pattern: it lags Growth portfolio (and S&P 500) during bull markets but has positive performance in four out of five bear markets we identify since 1970.

EXHIBIT 12

Goal-tilted Portfolio Weights



NOTES: Period: 1970–2022. Strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$. The four panels depict weights of eight assets in each Goal-tilted portfolio.

EXHIBIT 13**Goal-Tilted Portfolios Performance Statistics**

1970–2022	S&P	Bond Agg	Goal Parity Balanced	Growth	Income	Preservation	Liquidity
Mean, annualized	10.9%	6.7%	9.63%	11.6%	10.3%	9.4%	6.3%
Stdev, annualized	15.4%	5.3%	7.84%	12.8%	10.3%	7.8%	3.6%
Worst Month	–21%	–6%	–11%	–18%	–16%	–7%	–4%
Best Month	18%	11%	9%	18%	12%	9%	9%
Sharpe Ratio	0.43	0.44	0.68	0.57	0.58	0.65	0.56
Worst Drawdown	–51%	–17%	–22%	–39%	–39%	–13%	–9%
2002 Drawdown	–24%	–17%	–10%	–18%	–18%	–2%	–9%
2022 Return	–18%	–13%	–5%	–10%	–11%	9%	–7%
Bear Markets	S&P	Bond Agg	Goal Parity Balanced	Growth	Income	Preservation	Liquidity
1972.12–1974.09	–45%	7%	–4%	–6%	–24%	17%	6%
1987.08–1987.11	–29%	2%	–9%	–8%	–14%	–8%	0%
2000.08–2002.09	–45%	23%	–3%	–19%	–18%	21%	17%
2007.10–2009.02	–51%	6%	–22%	–39%	–39%	1%	2%
2021.12–2022.09	–24%	–15%	–10%	–18%	–18%	8%	–8%

NOTES: Period: 1970–2022. Strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$.

Interesting in their own right, our goal-tilted portfolio illustrations represent a “maximal” goal tilt to each of the four goals. A goal-oriented investor may instead choose to target a more moderate tilt (such as 50% growth, or 50% income) for the overall set of assets, which will give a result in between the Goal Parity Balanced, and the full 100%-tilt portfolios. Or, our investor may choose to efficiently incorporate a goal-tilted portfolio as an investment within the overall portfolio, for the purpose of “goal-completion” or “goal-balancing.”

Equities and Bonds

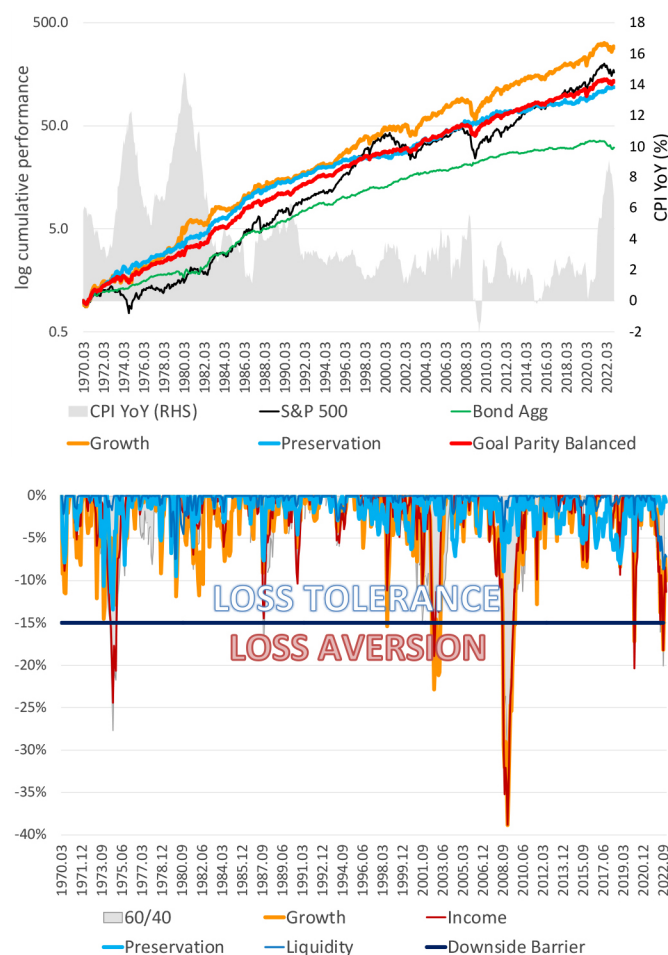
Observe that bonds had a much larger weight in the Goal Parity Balanced and Preservation portfolios in the late 1970s and 1980s. This is because bonds were much more volatile in that period, positively correlated with stocks and contributing more to the Growth and Preservation goals. In the 21st century negatively-correlated-with-stocks and much-less-volatile bonds have been contributing more to Liquidity and Income.

Investor-Specific “Left-Tail” Risk and Portfolio Drawdowns: Discussion

One of our investor-specific parameters is B , the “substantial loss” barrier. In our example here $B = 0.85$, meaning that the investor is very averse to losing more than $(1 - B) = 15\%$ of the portfolio’s net worth but tolerant to drawdowns of less than 15% in order take advantage of investment opportunities. Empirically, Exhibit 10 demonstrates that Goal Parity Balanced portfolio simulation breaches the 15% downside (from high watermark) barrier only once, briefly during the GFC, and Exhibit 14 demonstrates that Preservation and Liquidity portfolio simulations never breach the 15% downside barrier, although there are some close calls for the Preservation portfolio. In this sense, our goal-based asset allocation may be regarded as a drawdown-aware portfolio construction methodology, seeking to keep losses within the investor-specified tolerance level. The “substantial loss” barrier B may be related to investor’s liabilities and our framework may be used to complete LDI portfolios (see, for example, Wilcox, Horvitz, and diBartolomeo 2006).

EXHIBIT 14

Goal-Tilted Portfolios Cumulative Performance and Drawdowns



NOTES: Period: 1970–2022. Strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$. Upper panel depicts cumulative performance (log-scale y-axis). Lower panel depicts drawdowns from high watermark. Simulated performance is not indicative of actual future results. For illustration and educational purposes only.

Goal Parity Balanced: Varying T and B

So far, we have been working with strategic horizon $T = 10$ years and “substantial loss” barrier $B = 85\%$. How would the Goal Parity Balanced portfolios change with different investor’s preferences? First, we recall that varying T and B changes the GIPL Decomposition. For example, a $T = 3$ year horizon risk-averse investor should consider High Dividend Equities as mostly contributing to Growth and rely less on this asset for Income and Preservation, much less Liquidity. By contrast, longer-horizon risk-seeking investors can “source” Income and Liquidity from High Dividend Equities, and would need more assets like Small Cap and Private Equity to help achieve their Growth goal.

We illustrate the dependence of Goal Parity Balanced portfolios on T and B in Exhibit 15. Observe that for an investor with a strategic horizon $T = 3$ years, averse to losing beyond $B = 95\%$ of portfolio’s net value, the Goal Parity Balanced portfolio is mostly bonds, diversified with equities and CTAs. By contrast, for an investor with a strategic horizon $T = 30$ years, tolerant to drawdowns of less than $(1 - B) = 30\%$, the Goal Parity Balanced is a diversified growth portfolio of equities, gold, and CTAs. We report and depict simulated performance of the Goal Parity Balanced portfolios with T and B as above in Exhibits 16 and 17.

Endowment Portfolios

We performed the simulations above using eight assets commonly used and accessible by institutional and retail investors alike. We would like to touch upon an “Endowment Investment Model” which employs a wide array of assets including private equity, real estate, infrastructure, timberland and hedge funds cf. Ang et al. (2020) and Shepard, Qiu and Li (2022).

We illustrate Goal Parity Balanced portfolios with endowment-style assets in Exhibit 18. We form three portfolios in December 2021 for $T = 10$ years and three “substantial loss” barriers $B = 0.85$, $B = 0.80$, and $B = 0.70$. The more risk-averse portfolios with $B = 0.85$ sources Liquidity and Preservation from investment grade bonds and TIPS, while the more risk-seeking portfolios with $B = 0.70$ sources more Preservation from Commodity Producers and Global Macro. Sometimes, however, a greater appetite for differing risks does not necessarily mean more risk. Using 2022 performance as a case study, the risk-seeking portfolio performed better because bonds disappointed and Commodity Producers and Global Macro did so well.

Summarizing, we observe that endowments have access to a wider universe of assets, and may practically invest significant fractions of their wealth into less liquid asset classes. Further, they manage explicitly to a longer horizon. The Goal Parity Balanced portfolio uses the natural Endowment advantages above to allocate more capital to Growth-ier but less liquid assets, while at the same time seeking to achieve the Liquidity and Preservation goals as well.

EXHIBIT 15

Examples of Goal Parity Balanced Portfolios, Varying T and B

2021.12 Asset	Investor Preferences		
	$T = 3, B = 0.95$	$T = 10, B = 0.85$	$T = 30, B = 0.70$
Small Caps	1%	12%	27%
S&P	8%	16%	15%
HD Equities	8%	16%	14%
HY Bonds	28%	19%	0%
Bond Agg	30%	6%	0%
1–3 yr Bonds	7%	2%	0%
Gold	1%	12%	30%
CTA	18%	17%	14%
TOTAL	100%	100%	100%

EXHIBIT 16

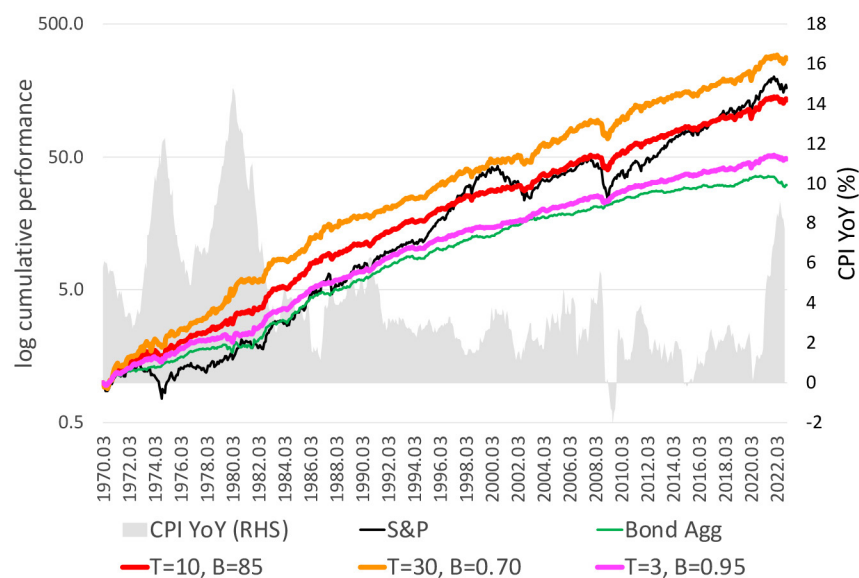
Goal Parity Balanced Simulated Portfolios, Varying T and B : Cumulative Performance

EXHIBIT 17

Goal Parity Balanced Simulated Portfolios, Varying T and B : Performance Statistics

1970–2022	S&P	Bond Agg	$T = 3, B = 0.95$	$T = 10, B = 0.85$	$T = 30, B = 0.70$
Mean, Annualized	10.9%	6.7%	7.5%	9.6%	11.3%
Stdev, Annualized	15.4%	5.3%	5.0%	7.8%	10.6%
Worst Month	–21%	–6%	–6%	–11%	–14%
Best Month	18%	11%	10%	9%	13%
Sharpe Ratio	0.43	0.44	0.64	0.68	0.66
Worst Drawdown	–51%	–17%	–12%	–22%	–28%
2002 Drawdown	–24%	–17%	–8%	–10%	–13%
2022 Return	–18%	–13%	–5.4%	–5.1%	–4.9%

NOTES: Period: 1970–2022. Strategic horizons $T = 3$, $T = 10$, $T = 30$ years, “substantial loss” barriers $B = 95\%$, $B = 85\%$, $B = 70\%$ respectively. Cumulative performance (log-scale y-axis) of Goal Parity Balanced portfolios. Simulated performance is not indicative of actual future results. For illustration and educational purposes only.

EXHIBIT 18

Examples of Goal Parity Balanced Portfolios for Endowments, Varying B

2021.12		Investor Preferences		
Asset	2022 Return	$T = 10, B = 0.85$	$T = 10, B = 0.80$	$T = 10, B = 0.70$
Private Equity (proxy)	-17%	7%	10%	19%
S&P 500	-18%	15%	17%	18%
REIT	-25%	11%	12%	13%
S&P Global Infra	0%	13%	15%	20%
Long TIPS	-32%	25%	22%	6%
Bond Agg	-13%	20%	13%	9%
Commodity Producers	22%	6%	6%	8%
Global Macro	55%	3%	4%	7%
TOTAL		100%	100%	100%
2022 Performance		-14%	-12%	-7%

NOTES: Endowment assets are represented by

- Private Equity: proxied by MSCI USA Small Cap Gross Total Return USD Index
- Global Infrastructure: S&P Global Infrastructure Total Return Index
- REIT: Dow Jones Equity REIT Total Return Index
- Commodity Producers: MSCI World Commodity Producer Sector-Capped Gross Total Return USD Index
- Global Macro: SG Trend Index, 2x-levered

STRATEGIC ASSET ALLOCATION AS TRACKING NOMINAL GDP AFTER CONSUMPTION

Cumulative return indexes usually assume compounded reinvestment of all returns, but in practice asset owners may have consumption needs and so may need to withdraw some money from their investment portfolios on a regular basis. Assuming annual consumption of 3% of the investment portfolio, we observe that, empirically, the following portfolios may be regarded as “tracking” US Nominal GDP, which includes both real growth and inflation:

- S&P 500 Index⁶ (the index itself, not total return)
- MSCI USA Total Return Index less 3% annual consumption
- 60% S&P/40% Bond Agg less 3% annual consumption
- Goal Parity Balanced $T = 10, B = 0.85$ less 3% annual consumption

If we consider assets to be claims on the nominal economy, and if we think about nominal GDP as an after-consumption Strategic Asset Allocation (Capital Allocation) *benchmark*, then we may view Strategic Asset Allocation as seeking to minimize the tracking error to nominal GDP (any alpha can be added on top of that). How can this be achieved? Using the 1970–2022 data we observe the following, with results shown in Exhibits 19 and 20.

- Stocks: empirically, S&P 500 “tracks” nominal GDP with 17% tracking error (annual data) and long waits to “get back” to nominal GDP.
- 60/40 = Stocks + Bonds: adding bonds cuts the pre-consumption drawdowns, e.g., from 51% to 33% in GFC, and cuts the tracking error (annual data) to 11%.

⁶S&P dividend yield averages about 3% since 1970: higher before 1995, lower after 1995.

EXHIBIT 19

Strategic Asset as Allocation Tracking Nominal GDP: Cumulative Performance

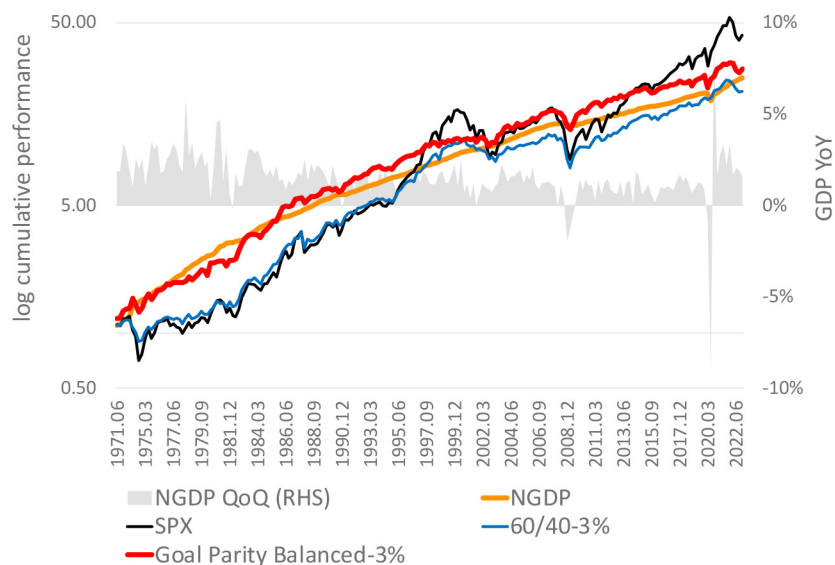


EXHIBIT 20

Strategic Asset as Allocation Tracking Nominal GDP: Performance Statistics

1970–2022, qtr.	NGDP	S&P	M2US–3%	60/40–3%	Goal Parity Balanced–3%
Mean, annualized	6.2%	8.6%	8.3%	6.3%	6.7%
Stdev, annualized	2.6%	16.6%	16.9%	9.9%	8.6%
Skewness	–1.31	–0.62	–0.60	–0.58	–0.11
Worst Quarter	–9%	–26%	–26%	–19%	–14%
Best Quarter	9%	22%	22%	15%	17%
NGDP Tracking Err.	0%	17%	17%	11%	8%

NOTES: Period: 1970–2022. The optimized returns presented are for a simulated portfolio. Simulated performance does not represent actual trading or actual trading account but rather were achieved by means of retroactive application of a 4x4invest’s proprietary asset allocation model in which expected returns are assumed to be zero, strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$. Optimized weights changed in December each year. Simulated performance is not indicative of actual future results.

But there are still long waits to “get back” to nominal GDP, especially with inflationary conditions.

- Goal Parity Balanced portfolios include Stocks and Bonds, and also built-in exposure to both high-yielding Income-contributing assets (that may sometimes be defensive in a crisis), and Preservation-contributing assets such as CTAs and commodities that may help mitigate the inflation-induced crises. Empirically, the pre-consumption GFC drawdown of the eight-asset Goal Parity Balanced is 22%. The nominal GDP tracking error is cut to 8%.

A similar relationship between asset allocation portfolios and GDP may be observed in other relatively closed developed markets such as Eurozone and Japan. However, publicly traded asset market performance may significantly diverge from official GDP growth numbers in smaller, more open, and emerging markets, perhaps in part due to global influences and lower securitization of economic activity in some of these markets.

CONCLUSION

We have illustrated our framework's quantification of the investor-specific Growth, Income, Preservation, and Liquidity goals, using data for a wide universe of public and private assets representing all major asset classes across several decades, as well as a global asset panel, "endowment" asset universe, and Fama-French factor portfolios in the Appendixes.

Further, we have simulated goal-based strategies going back to 1970, using eight assets that represent a universe with substantial contributions to all our four goals. Due to this goal diversification, the Goal Parity portfolio simulation exhibited returns comparable with equities with about half the volatility and much shallower drawdowns. The Goal Parity simulated performance is competitive with hedge fund indexes in terms of returns and drawdowns as well.

Our goal-based portfolios can be tilted to one or more goals. In particular, the eight-asset Growth simulated portfolio diversifies equities and thus exhibits higher returns and lower drawdowns. Our Preservation portfolio may be interesting in its own right: it exhibits a very different performance pattern, outperforming in a volatile or inflationary environment at the expense of somewhat lower returns in an equity bull market.

Moreover, we illustrated the effects of customizing the Goal Parity portfolios to investor-preferences: strategic horizon and loss tolerance. Our methodology delivers a more drawdown-resilient portfolio to a shorter-horizon, less loss-tolerant investor, at the expense of lower long-term returns, and a growth-ier portfolio to a longer-horizon, more loss-tolerant investor.

In practice, our goal-based portfolios would utilize a richer, more robust asset universe, as well as investor-defined positioning limits, cash flow requirements, and other constraints. This goal-based methodology also can be used to complement other widely used portfolio construction techniques, for example in order to goal-tilt or complete existing portfolios or strategies. One application is the completion of Liability-Driven Investment (LDI) portfolios, which by construction cover the Liquidity and Income goal in the LDI sense, to be more goal-balanced. Incorporating strategic expected returns (capital assumptions) with any forecasting power would improve investment performance. Overall, we have empirically illustrated a powerful goal-based, investor-specific asset management framework.

APPENDIX A

SKEWNESS AND HORIZON-DRIVEN LIQUIDITY PREFERENCE

Defaultity and Liquidity

Following Cron and Golts (2022), we provide a direct approach to estimating σ_D and σ_L for each asset from its volatility σ and skewness γ . We model the asset returns X as the sum of two factors:

$$X = \underbrace{X_{D0} + X_{D-}}_{\text{default factor } X_D} + \underbrace{X_{L0} + X_{L+}}_{\text{liquidity factor } X_L},$$

where X_D is the negatively skewed "default risk" factor reflecting deeper drawdown risk for cyclical, carry, high-income, productive assets, and X_L is the positively skewed "liquidity risk" factor, relevant for long-horizon, inflation-resistant, safe haven, and scarce assets. Intuitively, we model σ_D and σ_L , dubbed "defaultity" and "liquidity," as arising from (appropriately scaled) standard deviations of X_D and X_L respectively.

Further, each X_D and X_L is split into a symmetric, light-tail component (X_{L0} and X_{D0}) and a one-sided heavy tail component (X_{L+} with skewness $\gamma_{L+} > 0$, and X_{D-} with skewness $\gamma_{D-} < 0$). All four components are assumed to be zero-mean and independent for simplicity.⁷ We assume that the variance of each tail is a fixed fraction of each factor's variance:⁸

$$\rho^2 = \frac{\text{var}(\text{Tails})}{\sigma^2} = \frac{\text{var}(X_{D-})}{\text{var}(X_{D0}) + \text{var}(X_{D-})} = \frac{\text{var}(X_{L+})}{\text{var}(X_{L0}) + \text{var}(X_{L+})} \leq 1$$

Assuming $\gamma_{L+} = |\gamma_{D-}| = \gamma_{\text{tail}}$, we can derive the following result:

$$\frac{\sigma_L}{\sigma_D} \equiv \sqrt{\frac{1 + \frac{\gamma}{\gamma_{\text{tail}} \rho^3}}{1 - \frac{\gamma}{\gamma_{\text{tail}} \rho^3}}} \quad \text{thus} \quad \frac{\sigma_L}{\sigma_D} \approx \exp\left(\frac{\gamma}{\gamma_{\text{tail}} \rho^3}\right).$$

The exponential form is approximate for small γ , and gives a robust response for larger γ . Note that the ratio σ_L/σ_D depends only on γ and the product $\gamma_0 = \gamma_{\text{tail}} \rho^3$. Using the exponential form above, we normalize the defaultity and liquidityity as follows:

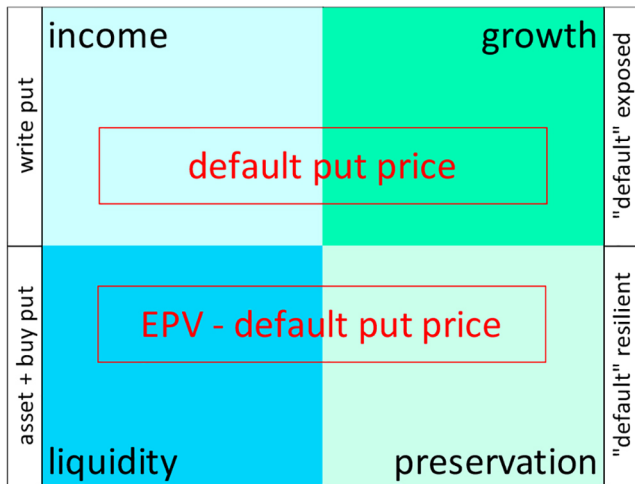
$$\sigma_L = \sigma \exp\left(\frac{\gamma}{2\gamma_0}\right) \quad \text{and} \quad \sigma_D = \sigma \exp\left(-\frac{\gamma}{2\gamma_0}\right).$$

Thus, if $\gamma < 0$ we have $\sigma_D > \sigma > \sigma_L$, meaning that the asset's "default risk" is greater than its "liquidity risk," and vice versa for $\gamma > 0$.

In practice, the empirical estimates of γ , σ_D and σ_L should be appropriately constrained.

EXHIBIT A1

Heuristic Investor-Specific Default Exposed/Resilient Asset Decomposition Scheme



Default Options

Let T be the strategic investment horizon and B be the "substantial loss" barrier for an investor. Following Golts-Kritzman (2010), we define the default (put) options as follows:

$$\Pi_{\text{default option}} = \Pi_{\text{first-passage}}(T, \sigma_D, B, \$1)$$

Intuitively, we can think about the following decomposition of an asset's Expected Present Value (EPV):

$$(\$1 - \underbrace{\Pi_{\text{default put}}(T, \sigma_D, B, \$1)}_{\text{"default" resilient}})EPV + \underbrace{\Pi_{\text{default put}}(T, \sigma_D, B, \$1)}_{\text{"default" exposed}}EPV$$

Liquidity Options: Horizon-Driven Liquidity Preference

Valuing liquidity—in particular using option pricing—is an effort to quantify the practical and common-sense relationship between assets' liquidity and investor's horizon, for example Hill (2009) observes that "Liquidity issues are most pronounced when an investor's horizon differs from that of the security being held." Kim (2014)

⁷ This overall construction is analogous to the sum of independent normal and exponential random variables producing an Exponentially Modified Gaussian distribution.

⁸ Given generally fat-tailed financial return distributions we can realistically expect $\rho^2 \geq 25\%$.

states: “We model the liquidity shock as the variability of holding period.” Golts and Kritzman (2010) proposed using a cliquet option as a liquidity option.

We describe here a liquidity option approach based on Bermudan options. More specifically, we price the investor-specific liquidity risk as follows. Introduce *one year* “liquidity strike” L_1 such that $0 \leq L_1 \leq 1$, and set *liquidity preference* $L = (L_1)^T \approx 1 - (1 - L_1)^T$. For example, if $L_1 = 0.985$ and $T = 10$ then

$$L = (0.985)^{10} = 0.8597,$$

that is, the investor is comfortable with keeping about $14\% = 100\% - 86\%$ of the portfolio’s *EPV* to be potentially not immediately accessible, while seeking to keep 86% of the portfolio’s *EPV* ready for immediate liquidation if needed. Conversely, if $L = 0.85$ and $T = 10$ then

$$L_1 = L^{1/T} = 0.85^{0.1} = 0.9839$$

Following Golts and Kritzman (2010), we subdivide the strategic investment horizon T into m equal length liquidity intervals $0 = T_0 < T_1 < \dots < T_m = T$. For example, $T = 10$ years and the *liquidity interval* $\tau = T_i - T_{i-1}$ is one quarter for any i , and $m = T/\tau = 40$. For a long-term investor holding some asset(s) on a strategic horizon T , having a “liquidity option” conceptually means that the investor can get some cash within (or at the end of) every liquidity interval⁹ τ . Our liquidity option here is a \$1-notional Bermudan option with exercise dates $T_1 < \dots < T_m = T$ and the pay-offs of $e^{rT_i}L$ if the barrier L is breached at T_i , and zero otherwise.

$$\Pi_{\text{liquidity option}} = \Pi_{\text{Bermudan}}(\tau, T, \sigma_L, L = (L_1)^T, \$1).$$

For example, if $\tau = T$ (no rebalancing), then our liquidity option is a European digital option. From now on we assume small $\tau \rightarrow 0$ (frequent rebalancing) and $L_1 = 0.985$. Then

$$\Pi_{\text{liquidity option}} = \Pi_{\text{first-passage}}(T, \sigma_L, L = (0.985)^T, \$1).$$

We call these liquidity options *horizon-driven* since their strikes are dependent¹⁰ on the strategic investment horizon T . An investor’s liquidity preference may range from, say, $L = 0.985$ for $T = 1$ to $L = (0.985)^{40} = 0.5463$ for $T = 40$. A saver with a 1-year horizon would seek to keep 98.5% of the portfolio liquidity resilient i.e., available for immediate liquidation, while an endowment with a 40-year horizon would seek to keep only 55% of the portfolio available for immediate liquidation and be comfortable with 45% of the portfolio dedicated to taking advantage of long-term growth and inflation protection opportunities.

Intuitively, we can think about the following decomposition of an asset’s *EPV*:

$$\begin{aligned} & \underbrace{(\$1 - \Pi_{\text{liquidity put}}(T, \sigma_L, L, \$1))}_{\text{liquidity resilient}} EPV + \underbrace{\Pi_{\text{liquidity put}}(T, \sigma_L, L, \$1)}_{\text{illiquidity exposed}} EPV \\ &= \underbrace{(\$1 - L \Pi_{\text{liquidity call}}(T, \sigma_L, 1/L, \$1))}_{\text{liquidity resilient}} EPV + \underbrace{L \Pi_{\text{liquidity call}}(T, \sigma_L, 1/L, \$1)}_{\text{illiquidity exposed}} EPV \end{aligned}$$

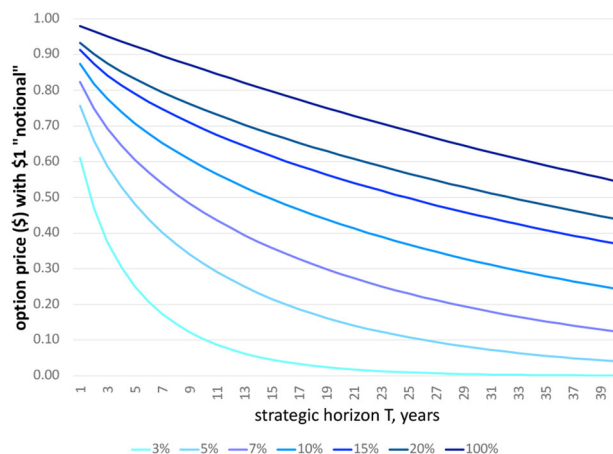
From the 4x4 Asset allocation perspective, if an investor’s horizon becomes shorter, then the liquidity options become more expensive, the portfolio becomes overweight in Growth and Preservation and needs to be rebalanced into more Liquidity and Income, and vice versa. Heuristically, these ideas are depicted in Exhibits A1 and A2. Exhibit A3 shows the liquidity option price across horizon, for various volatilities.

⁹ We interpret τ as the tactical rebalancing/trading frequency for the 4x4 Asset Allocation purposes.

¹⁰ These first-passage liquidity options no longer have a dependence on the tactical rebalancing frequency τ , see Golts and Kritzman (2010, formula 4).

EXHIBIT A2**Heuristic Illiquidity Exposed and Liquidity Resilient Asset Decomposition Scheme**

asset + write call	buy call
income	growth
EPV - liquidity call price	liquidity call price
liquidity	preservation
liquidity resilient	illiquidity exposed

EXHIBIT A3**Horizon-Driven Liquidity Option Prices vs Horizon T , Varying Volatility**

NOTES: Option tenor T varies between 1 and 40 years (x-axis), tactical rebalancing frequency $\tau \rightarrow 0$ (frequent rebalancing,) and one year “liquidity strike” $L_1 = 0.985$. The legend indicates volatility levels, for example the “3%” line depicts the liquidity option’s price dependence on the strategic horizon assuming 3% volatility. For simplicity we assume here that the “risk-free” rate $r_0 = 0$.

APPENDIX B**GIPL GOAL MAP**

Using the default and liquidity options, we map our universe of assets into the unit square

$$[0,1]^2 = [0,1] \times [0,1] = [0\%,100\%] \times [0\%,100\%],$$

where each Liquidity, Income, Preservation, and Growth goal corresponds to a corner of the square. Intuitively, the vertical dimension reflects the “defaultility” risk and the horizontal dimension reflects the “liquidity” risk. Since the substantial loss barrier B indicates where the investor’s utility turns sharply downward, the option pricing formula indicates what fraction of the asset’s value is “risk-relevant” or “at-risk” for the given investor. For example, an asset with 8% “defaultility” could be very risk-relevant for an investor with $B = 95\%$, and is markedly less risk-relevant for an investor with $B = 85\%$.

More precisely, denoting option prices arising from “defaultility” and “liquidity” risks¹¹ as

$$\Pi_D = \Pi_{\text{default option}} \quad \text{and} \quad \Pi_L = \Pi_{\text{liquidity option}}$$

respectively, we define what fraction of asset’s Expected Present Value (EPV) is risk-relevant along each of the “defaultility” and “liquidity” axes. The asset’s Growth, Income,

¹¹ See Appendix A for the definition of $\Pi_{\text{default option}}$ and $\Pi_{\text{liquidity option}}$.

Preservation, and Liquidity goal contributions—or fractions of the asset's *EPV*—are denoted *epvg*, *epvi*, *epvp*, *epvl* respectively, and are defined as follows:

Income = $epvi = \Pi_D \times (1 - \Pi_L)$	Growth = $epvg = \Pi_D \times \Pi_L$
Liquidity = $epvl = (1 - \Pi_D) \times (1 - \Pi_L)$	Preservation = $epvp = (1 - \Pi_D) \times \Pi_L$

The Asset Map is a visualization of the asset universe in the unit square $[0, 1] \times [0, 1]$: every asset is represented as a dot with coordinates (x, y) :

$$x = epvp + epvg = \Pi_L \quad \text{and} \quad y = epvi + epvg = \Pi_D.$$

Since $epvg + epvi + epvp + epvl = 1$, such (x, y) always lies in the unit square $[0, 1] \times [0, 1]$.

For example, an asset is in the upper half of the square (the Income and Growth quadrants) if $\Pi_D > 50\%$ indicating significant “defaultility” risk for the investor; otherwise it is in the lower half. Similarly, an asset is in the right half of the square (the Growth and Preservation quadrants) if $\Pi_L > 50\%$, indicating significant “liquidity” risk for the strategic horizon T investor. Finally, each asset in the square may be understood geometrically: its location is precisely the goal-weighted average of the four unit square corners.

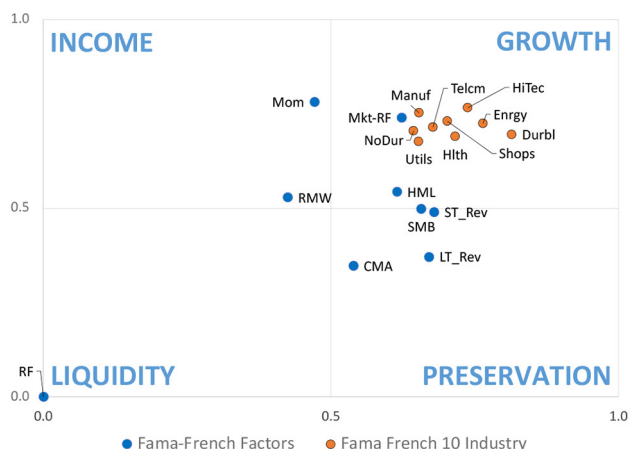
APPENDIX C

FAMA-FRENCH PORTFOLIOS

In this section we consider a universe of “Fama-French” assets and factors: equity long-only and long-short portfolios available at [Ken French's website](#). Using the data over the period 1963.07–2022.12 and following the GIPL Asset Map methodology described in **Appendix B**, we visually depict the assets' goal decomposition in Exhibit C1. We observe that over the entire almost 60-year period, the largest Preservation contribution is provided by CMA (Conservative Minus Aggressive) and Long-Term Reversal factors. The correlations of the factors are shown in Exhibit C2.

EXHIBIT C1

GIPL Asset Map: Investor's Goals: Fama-French



NOTES: 4x4invest's proprietary methodology, strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$. Period: 1963.07–2022.12.

EXHIBIT C2

Fama-French Correlations

Assets	Mkt-RF	SMB	HML	RMW	CMA	RF	Mom	ST_Rev	LT_Rev
Mkt-Rf	100.00%	27.95%	-20.92%	-18.03%	-36.57%	-8.48%	-16.46%	31.22%	-0.18%
SMB	27.95%	100.00%	-2.59%	-34.83%	-9.69%	-3.57%	-6.13%	17.73%	31.71%
HML	-20.92%	-2.59%	100.00%	9.05%	68.27%	6.69%	-20.41%	1.93%	50.85%
RMW	-18.03%	-34.83%	9.05%	100.00%	-1.91%	-1.04%	8.04%	-8.57%	-26.76%
CMA	-36.57%	-9.69%	68.27%	-1.91%	100.00%	5.57%	-2.11%	-13.36%	51.50%
RF	-8.48%	-3.57%	6.69%	-1.04%	5.57%	100.00%	6.79%	3.90%	1.25%
Mom	-16.46%	-6.13%	-20.41%	8.04%	-2.11%	6.79%	100.00%	-30.74%	-9.53%
ST_Rev	31.22%	17.73%	1.93%	-8.57%	-13.36%	3.90%	-30.74%	100.00%	9.04%
LT_Rev	-0.18%	31.71%	50.85%	-26.76%	51.50%	1.25%	-9.53%	9.04%	100.00%

NOTES: Period: 1963.07–2022.12. Standard Pearson correlation coefficients computed with monthly returns.

APPENDIX D

DATA DETAILS

We provide the details and summary statistics for our universe of (mostly) US-centric assets from March 2000 to December 2022. Data include the S&P 500 and Global Infrastructure Indexes, MSCI US Equity Indexes, Nasdaq 100 Index, Dow Jones Equity REIT Index, Bloomberg Fixed Income Indexes, HFRI Indexes, CTA Indexes from Barclay-Hedge, SG, DB, long and short volatility indexes from Eurekahedge/CBOE and DB, and Bloomberg Private Equity Indexes, see Exhibits D1 and D2. All data is monthly, except Private Equity indexes data is quarterly.

EXHIBIT D1

Decomposition of Asset Contributions to the Four Goals, 2000–2022

Assets					Decomposition of the Asset Contributions to the Four Goals			
Short Name	Description	Return	Volatility	Skew	Liquidity	Preservation	Income	Growth
S&P	S&P 500 Total Return Index	7.7%	15.5%	−0.50	10%	16%	28%	46%
Value	MSCI USA Value Gross Total Return USD Index	7.6%	15.5%	−0.55	10%	16%	29%	45%
Growth	MSCI USA Growth Gross Total Return USD Index	7.3%	17.3%	−0.46	9%	16%	26%	49%
MSCI US	MSCI USA Gross Total Return USD Index	7.5%	15.6%	−0.49	10%	16%	28%	46%
Quality	MSCI USA Quality Gross Total Return USD Index	8.5%	14.7%	−0.41	11%	17%	27%	45%
Nasdaq	Nasdaq-100 Total Return Index	7.7%	23.3%	−0.59	6%	14%	23%	56%
Small Caps	MSCI USA Small Cap Gross Total Return USD Index	10.0%	19.9%	−0.48	7%	16%	24%	53%
HD Equities	MSCI USA IMI High Dividend Yield Gross Total Return USD Index	9.8%	13.8%	−0.67	12%	14%	33%	41%
Min Vol Equities	MSCI USA Minimum Volatility Gross Total Return USD Index	8.6%	12.1%	−0.70	14%	14%	36%	36%
Global Infrastructure	S&P Global Infrastructure Total Return Index	9.9%	15.5%	−1.19	11%	10%	40%	39%
REITs	Dow Jones Equity RE IT Total Return Index	12.0%	20.6%	−0.81	7%	13%	28%	52%
Natural Resources	S&P Global Natural Resources Total Return Index	11.0%	20.8%	−0.34	7%	17%	22%	55%
Treasuries	Bloomberg US Treasury Total Return Index	3.6%	4.6%	−0.09	55%	18%	20%	7%
Long Treasuries	Bloomberg Long US Treasury with 10+ Y Total Return Index	5.6%	11.3%	0.18	15%	28%	20%	37%
IG Credit	Bloomberg US Corp Investment Grade Total Return Index	4.9%	6.1%	−0.82	36%	8%	46%	10%
Bond Agg	Bloomberg US Agg Total Return Index	4.0%	3.9%	−0.58	62%	5%	31%	2%
Long Bonds	Bloomberg 10+ yr Gov/Credit Total Return Index	6.0%	9.7%	−0.12	19%	24%	26%	32%
HY Bonds	Bloomberg US Corp High Yield Total Return Index	6.5%	9.4%	−1.10	20%	8%	51%	21%
TIPS	Bloomberg US Treasury Inflation-Linked Total Return Index	5.0%	5.9%	−0.85	37%	7%	47%	9%
Long TIPS	Bloomberg US Treasury Inflation Notes 10+ Y Total Return Index	6.2%	11.2%	−0.63	16%	15%	36%	34%

(continued)

EXHIBIT D1 *(continued)***Decomposition of Asset Contributions to the Four Goals, 2000–2022**

Assets					Decomposition of the Asset Contributions to the Four Goals			
Short Name	Description	Return	Volatility	Skew	Liquidity	Preservation	Income	Growth
1–3 yr Bonds	Bloomberg 1–3 yr Gov/Credit Total Return Index	2.7%	1.5%	0.16	99.7%	0%	0%	0%
T-Bill	Bloomberg US Treasury Bill Index	1.6%	0.5%	1.21	100.0%	0%	0%	0%
Gold	Gold Spot price per Troy Ounce	9.4%	16.3%	0.04	9%	23%	20%	49%
DXY	US Dollar Index	0.2%	7.9%	0.13	26%	30%	21%	23%
CBOE PutWrite	CBOE S&P 500 PutWrite Index	6.4%	11.1%	–1.65	16%	9%	48%	27%
DB Vol Carry	Deutsche Bank Volatility Carry (US Large Cap)	6.5%	4.9%	–1.01	44%	3%	50%	3%
HF Merger Arb	HFRI ED Merger Arbitrage Index	5.0%	4.0%	–2.22	51%	1%	47%	1%
HF Distressed	HFRI ED Distressed/Restructuring Index	6.7%	6.5%	–1.38	32%	5%	55%	8%
HF Rel Val	HFRI Relative Value Total Index	5.7%	4.5%	–3.39	47%	1%	50%	2%
HF Comp	HFRI Fund Weighted Composite Index	5.1%	6.3%	–0.87	34%	7%	48%	11%
HF Equity	HFRI Equity Hedge Total Index	5.0%	8.6%	–0.69	23%	13%	41%	23%
HF EM	HFRI Emerging Markets Total Index	6.4%	11.1%	–0.89	16%	12%	42%	30%
HF Macro	HFRI Macro (Total) Index	4.6%	5.1%	0.31	48%	32%	12%	8%
HF Defensive	HFRI FOF Market Defensive Index	4.0%	4.6%	0.15	54%	26%	14%	6%
BarHedge CTA	BarclayHedge CTA Index	3.7%	6.0%	0.38	38%	36%	13%	12%
SG CTA	SG CTA Index	5.0%	8.7%	0.18	23%	31%	20%	27%
SG Trend	SG Trend Index	6.9%	13.7%	0.16	12%	26%	19%	43%
DB CTA	DB Cross Asset CTA Trend Index	4.1%	5.5%	0.28	43%	33%	14%	11%
Eureka CBOE Long Vol	Eurekahedge CBOE Long Volatility Index	3.6%	7.4%	3.87	26%	58%	5%	10%
DB US Rates Vol	Deutsche Bank USD Rates Long Volatility Index	1.6%	2.7%	0.33	85%	13%	1%	0%
DB EU Rates Vol	Deutsche Bank EUR Rates Long Volatility Index	1.5%	2.7%	1.16	65%	35%	0%	0%

NOTES: 4x4invest’s proprietary methodology, strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$. Period: 2000–2022, except: period for Global Infrastructure: 2001–2022, Natural Resources: 2002–2022.

EXHIBIT D2**Decomposition of Private Equity Contributions to the Four Goals, 2007–2022**

Assets					Decomposition of the Asset Contributions to the Four Goals			
Short Name	Description	Return	Adj. Vol.	Skew	Liquidity	Preservation	Income	Growth
PE All	Bloomberg Private Capital Index	10.1%	26.3%	–1.78	6%	11%	28%	55%
PE Buyout	Bloomberg Buyout Private Equity Index	11.6%	32.8%	–1.54	5%	11%	24%	60%
PE Debt	Bloomberg Debt Private Equity Index	8.4%	32.0%	–1.50	5%	11%	25%	59%
PE Real Estate	Bloomberg Real Estate Private Equity Index	7.2%	25.9%	–1.27	6%	11%	28%	54%
PE Real Assets	Bloomberg Real Assets Private Equity Index	7.8%	31.4%	0.13	4%	18%	15%	63%
PE Venture	Bloomberg Venture Capital Index	10.9%	34.4%	–0.03	4%	16%	15%	65%

NOTES: 4x4invest’s proprietary methodology, strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$. Period: Jun, 2007–December 2022. Quarterly data.

Skewness estimation is subject to statistical error, for a normally distributed sample of size n :

$$\text{var}(\text{sample skewness}) = \frac{6n(n-1)}{(n-2)(n+1)(n+3)}.$$

For our 2000–2022 sample, $n = 274$ months, and the skewness standard error ≈ 0.15 . In particular, skewness estimates of all equity indexes above are in excess of two standard errors.

Global Assets

Complementing the mostly US-centric universe above, we look at the global assets. Data include the MSCI Global Equity Indexes, Bloomberg Global Fixed Income Indexes (local return), FX Total Return Long Indexes from Bloomberg, FX Carry Strategy Return Indexes from S&P, Bloomberg, DB, GSAM, SG, the Bloomberg Commodities Total Return Index, and Bitcoin (BTCUSD) spot prices, see Exhibit D3. Period: September 2000 to December 2022.

EXHIBIT D3

Decomposition of Global Asset Contributions to the Four Goals, 2000–2022

Assets		Decomposition of the Asset Contributions to the Four Goals						
Short Name	Description	Return	Volatility	Skew	Liquidity	Preservation	Income	Growth
AU Equities	MSCI Australia Gross Total Return Local Index	8.8%	13.6%	−0.99	12%	11%	40%	36%
CA Equities	MSCI Canada Gross Total Return Local Index	6.9%	14.3%	−0.84	11%	13%	36%	40%
UK Equities	MSCI UK Gross Total Return Local Index	5.5%	13.5%	−0.55	12%	16%	31%	41%
EU Equities	MSCI EU Gross Total Return Local Index	4.8%	15.8%	−0.50	10%	16%	28%	46%
JP Equities	MSCI Japan Gross Total Return Local Index	3.9%	17.3%	−0.34	8%	18%	24%	50%
EM Equities	MSCI EM Gross Total Return Local Index	8.5%	17.0%	−0.47	9%	16%	26%	49%
US Equities	MSCI USA Gross Total Return Local Index	7.5%	15.6%	−0.49	10%	16%	28%	46%
BR Equities	MSCI Brazil Gross Total Return Local Index	14.1%	23.1%	−0.38	6%	16%	21%	57%
CN Equities	MSCI China Gross Total Return Local USD Index	9.9%	26.2%	−0.18	5%	17%	18%	60%
IN Equities	MSCI India Gross Total Return Local Index	10.5%	24.2%	−0.67	6%	14%	24%	57%
HK Equities	MSCI Hong Kong Gross Total Return Local Index	7.4%	20.8%	−0.10	7%	19%	19%	55%
KR Equities	MSCI Emerging Korea Gross Total Return Local Index	9.8%	22.4%	0.08	6%	20%	17%	57%
ZA Equities	MSCI Emerging South Africa Gross Total Return Local Index	12.6%	17.0%	−0.04	9%	21%	20%	50%
TR Equities	MSCI Emerging Turkey Gross Total Return Local Index	22.1%	35.2%	0.59	4%	20%	12%	64%
TW Equities	MSCI Emerging Taiwan Gross Total Return Local Index	6.8%	21.9%	0.09	6%	20%	17%	56%
AU Bond Agg	Bloomberg Global Agg–Australia Total Return Index Unhedged AUD	4.2%	5.2%	−0.01	47%	23%	20%	10%
CA Bond Agg	Bloomberg Global Agg–Canadian Total Return Index Unhedged CAD	4.2%	4.3%	0.03	60%	20%	15%	5%
JP Bond Agg	Bloomberg Global Agg–Japanese Total Return Index Unhedged JPY	1.3%	1.9%	−0.27	97%	0%	2%	0%
EU Bond Agg	Bloomberg EuroAgg Index	3.3%	4.0%	−0.73	59%	3%	36%	2%
US Bond Agg	Bloomberg US Agg Total Return Index	4.0%	3.9%	−0.58	62%	5%	31%	2%
AUD	AUDUSD Total Return Long–Long AUD, Short USD	5.0%	12.4%	−0.28	13%	20%	27%	40%
CAD	CADUSD Total Return Long–Long CAD, Short USD	2.8%	8.8%	−0.36	22%	19%	32%	27%
CHF	CHFUSD Total Return Long–Long CHF, Short USD	3.5%	10.0%	0.34	18%	33%	17%	31%
EUR	EURUSD Total Return Long–Long EUR, Short USD	2.3%	9.7%	0.03	19%	27%	23%	31%
GBP	GBPUSD Total Return Long–Long GBP, Short USD	1.6%	8.7%	−0.17	23%	23%	27%	27%
JPY	JPYUSD Total Return Long–Long JPY, Short USD	−0.2%	9.5%	−0.03	20%	25%	24%	31%
NOK	NOKUSD Total Return Long–Long NOK, Short USD	2.8%	11.7%	−0.19	15%	22%	26%	38%
NZD	NZDUSD Total Return Long–Long NZD, Short USD	6.1%	13.0%	−0.18	12%	21%	25%	42%
SEK	SEKUSD Total Return Long–Long SEK, Short USD	1.4%	11.2%	0.11	16%	27%	21%	36%
BRL	BRLUSD Total Return Long–Long BRL, Short USD	7.5%	17.1%	−0.38	9%	17%	25%	49%
TRY	TRYUSD Total Return Long–Long TRY, Short USD	6.8%	17.8%	−1.37	9%	11%	36%	44%
MXN	MXNUSD Total Return Long–Long MXN, Short USD	4.3%	11.0%	−1.00	16%	10%	45%	28%
ZAR	ZARUSD Total Return Long–Long ZAR, Short USD	4.5%	16.3%	−0.26	9%	19%	24%	48%
CNY	CNYUSD Total Return Long–Long CNY, Short USD	3.0%	3.1%	−0.47	77%	3%	20%	1%
KRW	KRWUSD Total Return Long–Long KRW, Short USD	2.6%	10.5%	0.08	17%	27%	22%	34%
S&P G10 FX Carry	S&P Risk Premia FX Carry G10 Index USD Total Return	2.4%	9.3%	−1.01	20%	9%	49%	22%
DB G10 FX Carry	DB CoreSeries FX Carry G10 USD Index	1.0%	8.8%	−1.10	22%	8%	52%	18%
Bloomberg G10 FX Carry	Bloomberg Cumulative FX Carry Trade Index for G10	1.3%	9.0%	−0.67	22%	14%	40%	25%
GSAM FX Carry	Bloomberg GSAM Fx Carry Index	1.9%	7.5%	−1.00	27%	7%	51%	14%
SG G10 FX Carry	SGI FX-G10 Carry Trade	−0.2%	6.9%	−0.46	32%	15%	36%	17%
Gold	Gold Spot price per Troy Ounce	9.4%	16.3%	0.04	9%	23%	20%	49%
BCOM	Bloomberg Commodity Index Total Return	9.0%	15.9%	0.85	10%	36%	11%	42%
Bitcoin	XBTUSD Spot Exchange Rate–Price of 1 XBT in USD	192.8%	193.6%	4.55	2%	13%	12%	73%

NOTES: 4×4invest’s proprietary methodology, strategic horizon $T = 10$ years, “substantial loss” barrier $B = 85\%$. Period: September 2000–December 2022, except: period for BCOM: 1960–2022, period for BTCUSD: 2010–2022.

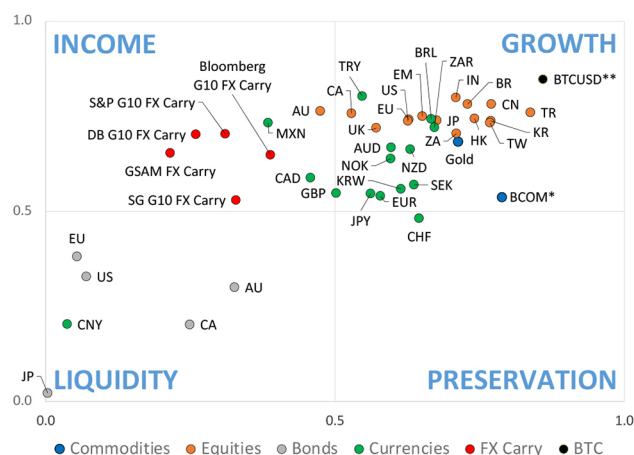
Following the methodology described in the Appendix B: GIPL Goal Map, we visually depict the assets' goal decomposition in Exhibit D4.

Data Used for the Simulations with Eight Traditional and Alternative Assets from March 1970 to December 2022

Data: S&P 500 Index, MSCI US Equity Indexes, Bloomberg Fixed Income Indexes, SG and BarclayHedge CTA Indexes. Index data are sourced from Bloomberg and deeper history from [Robert Shiller](#) and [Ken French](#). Author's proprietary methodology is used to simulate a CTA strategy in 1970–1980. All calculations and simulations are performed with monthly data. Monthly time series used for the simulations with traditional and alternative assets from March 1970 to December 2022 are reported in Exhibit D5.

EXHIBIT D4

GIPL Asset Map: Investor's Goals: Global Assets



NOTES: 4x4invest's proprietary methodology, strategic horizon $T = 10$ years, "substantial loss" barrier $B = 85\%$. Period: September 2000–December 2022, except period for BCOM: 1960–2022, period for BTCUSD: 2010–2022.

EXHIBIT D5

Monthly Time Series Used for Simulations with Traditional and Alternative Assets, March 1970 to December 2022

Short Name	Description
Small Caps	MSCI USA Small Cap Gross Total Return USD Index, extended before 1998 with Fama-French Small Cap (Lo 30) monthly returns index available via Ken French.
S&P	S&P 500 Total Return Index, extended before 1988 with MSCI USA Gross Total Return USD Index and Robert Shiller S&P data.
HD Equities	MSCI USA IMI High Dividend Yield Gross Total Return USD Index, extended before 1998 with Fama-French BIG HiDP monthly returns index available via Ken French.
HY Bonds	Bloomberg US Corp High Yield Total Return Index, extended before 1983 with Bloomberg US Corp Investment Grade Total Return Index and Robert Shiller bond data.
Bond Agg	Bloomberg US Agg Total Return Index, extended before 1976 with Bloomberg US Treasury Total Return Index and Robert Shiller bond data.
1–3 yr Bonds	Bloomberg 1–3 yr Gov/Credit Total Return Index, extended before 1976 with 1-mo T-bill data available via Ken French.
Gold	Gold Spot price per Troy Ounce.
CTA	SG CTA Index, extended before 1999 with BarclayHedge CTA Index and extended before 1980 with 4x4invest's Basic Short-Term Momentum Strategy.

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